

Electricity Consumption Forecasting

Introduction

This project focuses on predicting future electricity consumption using machine learning techniques based on historical data.

Objective

The objective is to accurately forecast electricity usage and help in efficient energy management by balancing supply and demand.

Dataset Description

The dataset includes features such as date, time, temperature, humidity, and day type, which influence electricity consumption.

Methodology

Data preprocessing was performed to clean the dataset and format date and time. Exploratory Data Analysis (EDA) was conducted to identify patterns and trends. Machine learning models such as Linear Regression, Decision Tree, and Random Forest were used.

Model Evaluation

Models were evaluated using metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Random Forest performed best.

Results

The model predicted electricity consumption patterns and showed that usage is highest during evening hours and lowest during midnight.

Conclusion

The project successfully demonstrated electricity consumption forecasting. Random Forest provided the most accurate predictions, and time of day was a key influencing factor.